

RemDrumStudio

TECHNICAL DESIGN DOCUMENT

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Revision History

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10/18/26	Justin Ericson Marcial	Exception Handling
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Functional Design Overview

Design Overview

This section shows how RemDrumStudio is designed to work. It explains the three main features of the system, namely Role-Based Authorization, Registration and Rescheduling Management, and Payment Tracking. Each feature has a flowchart that follows the steps of the Tutor, the Student and the System, from the first request to the final result.

1. Role-Based Authorization

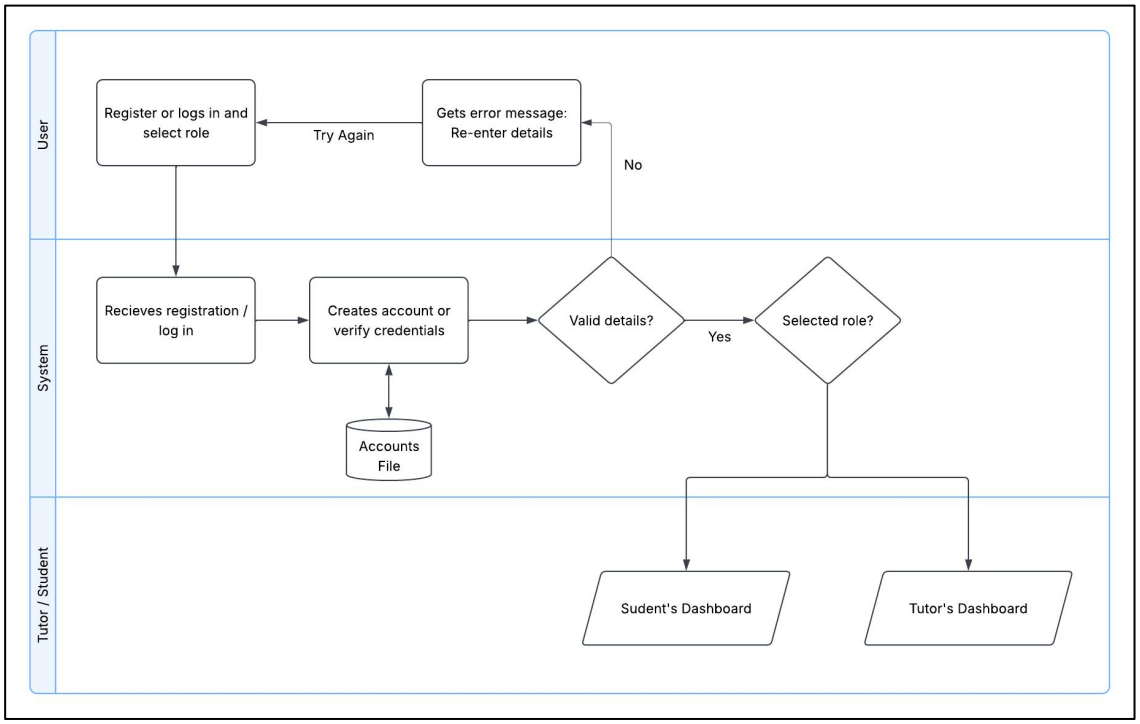


Figure 1. Role-Based Authorization Overview Design

Figure 1 shows how a user gains access to the system and how the system limits each user to the functions of their role. The user registers or logs in and selects a role, either student or tutor. The system creates an account for a new user, or checks the credentials of an existing user against the accounts file. If the details are invalid, it shows an error message and the user tries again. If they are valid, it opens the student dashboard or the tutor dashboard, depending on the selected role.

2. Registration and Rescheduling Management

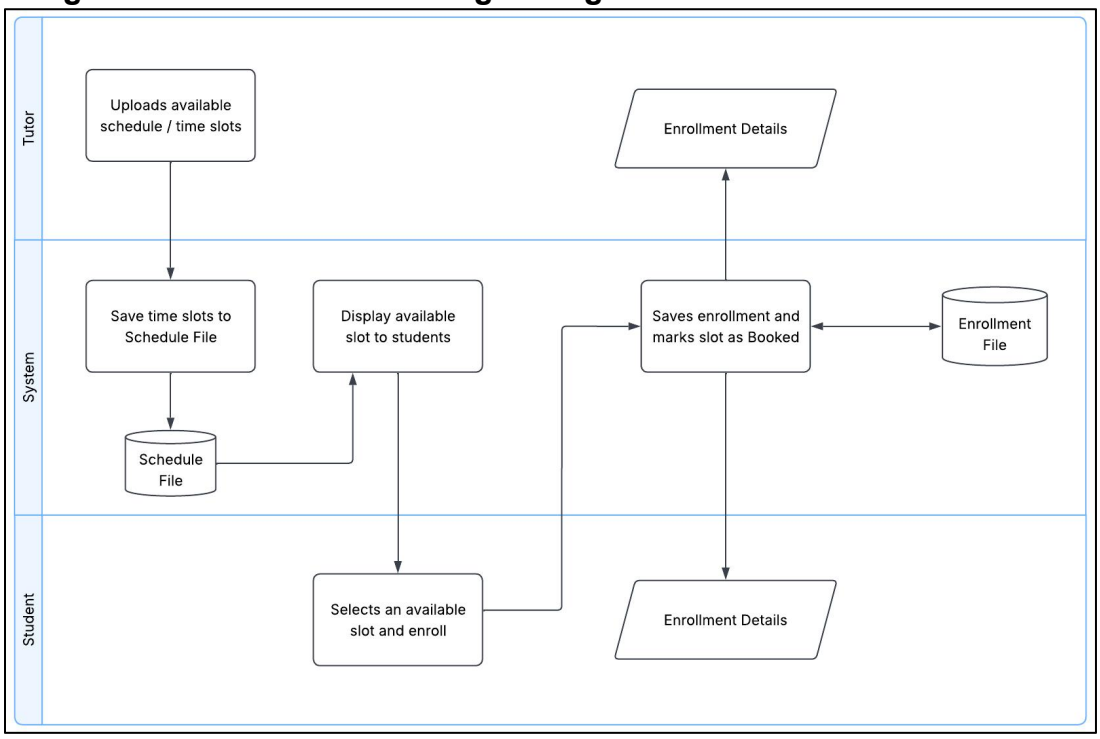


Figure 2. Registration Overview Design

Figure 2 shows how a session is made available and how a student books it. The tutor uploads the available time slots, and the system saves them to the schedule file and displays them to students. A student selects a slot and enrolls. The system saves the enrollment to the enrollment file, marks the slot as booked so no one else can take it, and sends the enrollment details to both the tutor and the student.

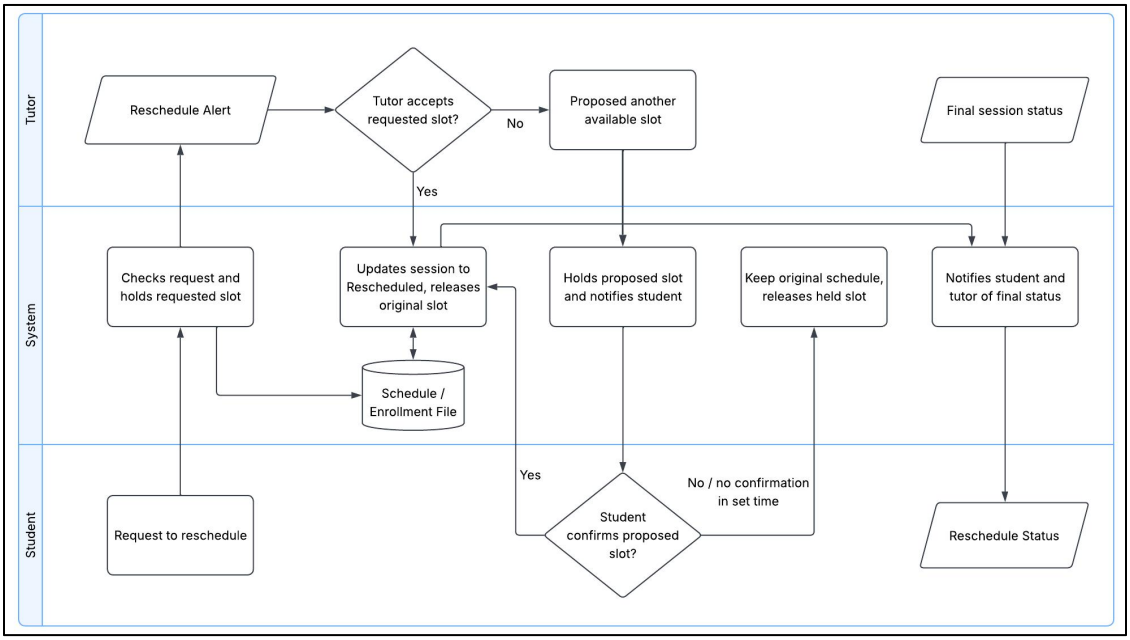


Figure 3. Rescheduling Overview Design

Figure 3 shows how a booked session is moved to a new time. The student requests a new slot and gives a reason, at least one day before the session. The system holds the requested slot, sets the session to pending confirmation and alerts the tutor. If the tutor accepts, the system updates the session to rescheduled and releases the original slot. If not, the tutor proposes another slot, which the system holds and sends to the student. If the student confirms, the session is updated to rescheduled. If the student declines or does not reply in time, the original schedule stays and the held slot is released. In every case, the system notifies both the student and the tutor of the final status.

3. Payment Tracking

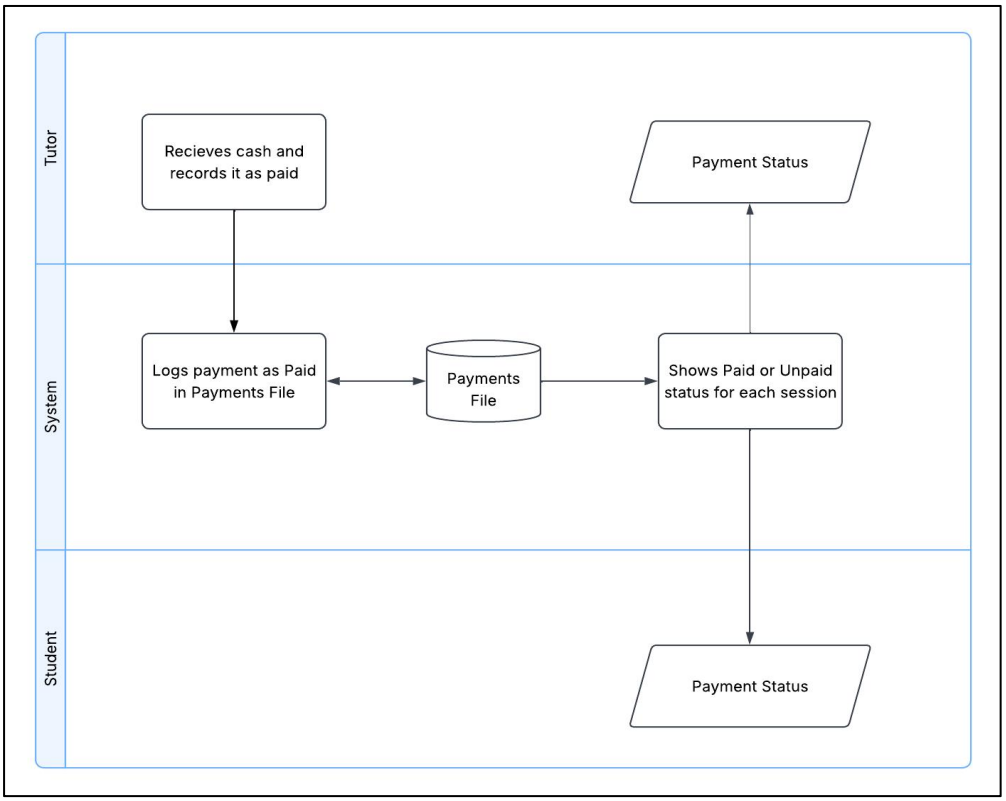


Figure 4. Payment Tracking Overview Design

Figure 4 shows how the payment for a session is made and recorded. The student pays the lesson fee in cash, and the tutor receives the cash and records the session as paid in the system. The system logs the session as paid in the payments file, shows the paid or unpaid status of each session, and send the payment status to both the student and the tutor.

Assumptions

The following assumptions were identified during the design phase.

1. Users can access the system through a web browser on a phone or a laptop with a stable internet connection.
2. Users can only access features that are allowed for their assigned roles.
3. Users provide complete and accurate information when registering.
4. Tutors can upload their available time slots.
5. Lessons are one-on-one, so each time slot can be booked by only one student.
6. A reschedule request is only be accepted as reschedule if the tutor will confirm it.
7. The tutor can approve or change payment and rescheduling records.

Technical Solution

Components Overview

This section lists the components that will be delivered for RemDrumStudio. It identifies the database and its tables, the application system, and the script used to build it. The following components will be delivered as part of this project.

No.	Component	Description
1	1 Database	Using MySQL, RemDrumStudio utilizes one database that stores the accounts, schedules, enrollments, reschedule requests, payments of the system.
2	6 Database Tables	6 database tables are identified that will hold the different data of the system (accounts, students, tutors, schedule slots, enrollments, and payments).
3	1 Application System	A web-based system used through a browser on a phone or a laptop, with a Student Dashboard and a Tutor Dashboard.
4	JavaScript, HTML, and CSS	The system is build using HTML for the page structure, CSS for the design and layout, and JavaScript for the system function.

Table 1. Components Overview

User Interface Design

This section presents the user interface of RemDrumStudio, which is the set of pages that the users see and use. It starts with the landing page, where a user signs up or logs in, followed by the student dashboard and tutor dashboard, which provide the pages and functions of each role.

A. Landing Page

The landing page serves as the main page of the RemDrumStudio website. It displays the website logo, navigation menu, and a background image showing people playing musical instruments. It also includes a short introduction and a Book a Session button that allows users to start booking a music session. The page helps users understand the purpose of the website and easily access its main features.

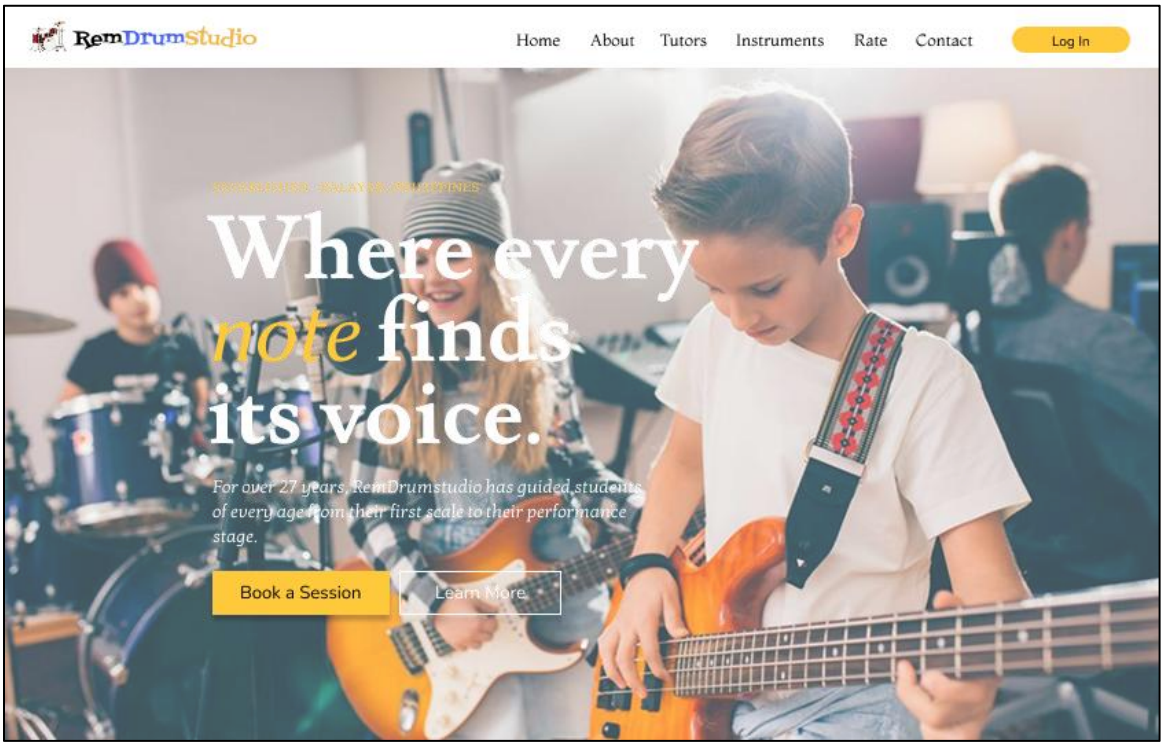


Figure 5. Landing Page of RemDrumStudio

While in figure 6, the log in page, allows registered users to access their accounts using their email address and password. It also provides a Forgot Password option for users who need to recover their accounts. The page includes a sign-up link for new users who do not have an account yet. Its main purpose is to provide users with a simple and organized way to access the website.

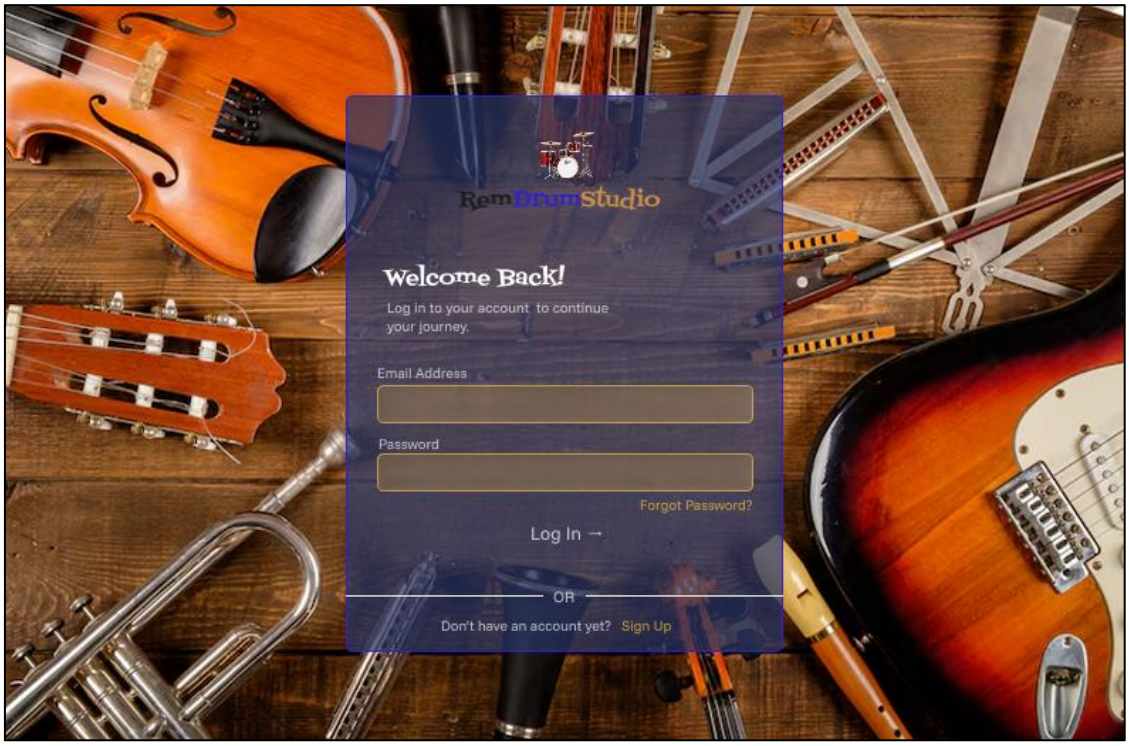


Figure 6. Log In Page

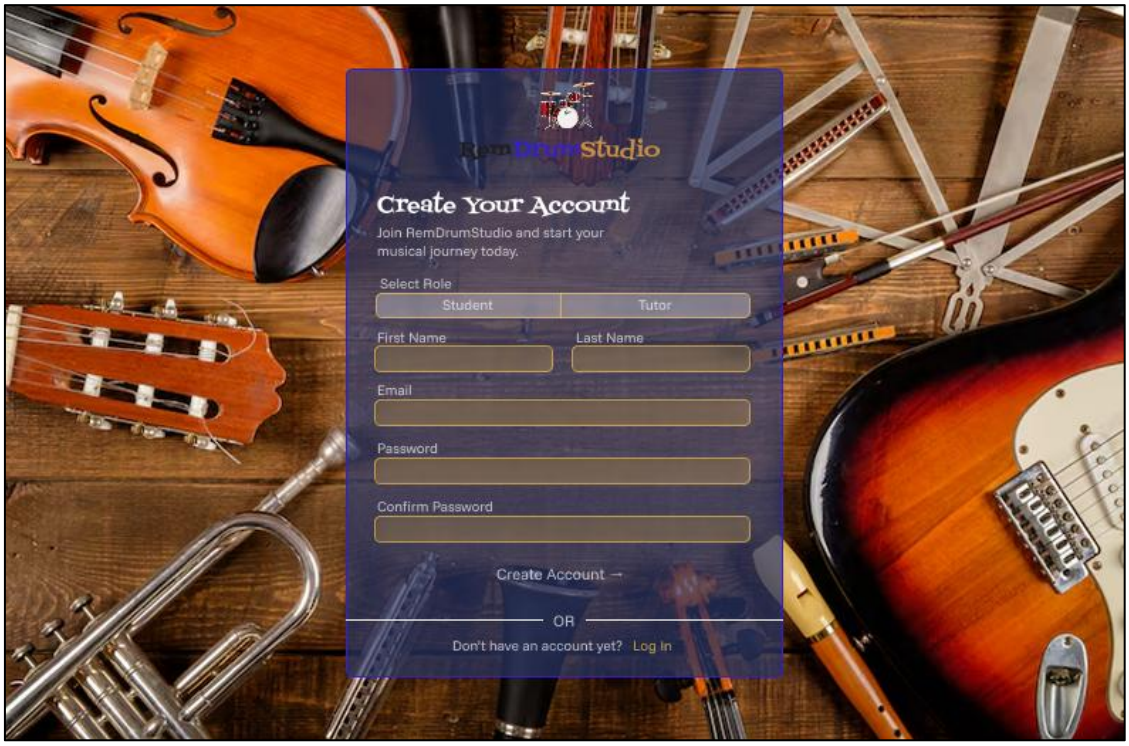


Figure 7. Sign Up Page

Lastly, the sign-up page allows new users to create an account on the RemDrumStudio website. It provides an entry point for users to register and begin using the website's services. The page also includes a link to the log in page for users who already have an account. This feature helps organize user access and allows new users to become part of the system.

B. Tutor's Dashboard

The tutor dashboard is the workspace of the tutor in RemDrumStudio. It allows the tutor to manage available lesson times, view students and bookings, respond to reschedule requests, and record cash payments. The following pages are available to the tutor.

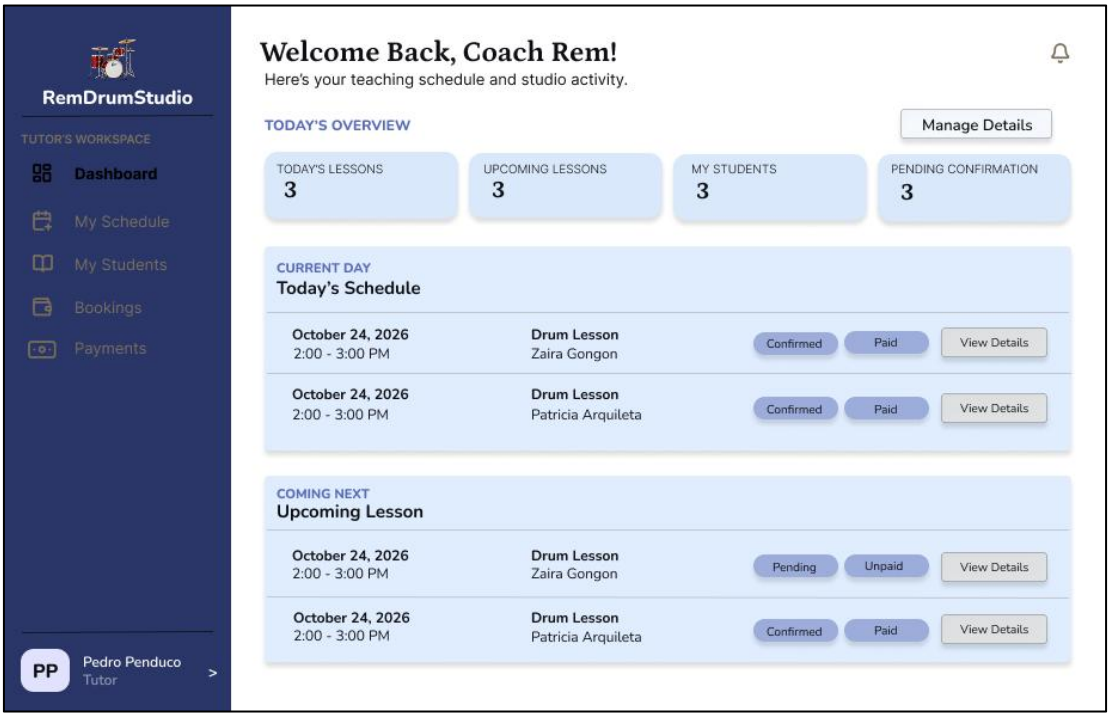


Figure 8. Tutor's Dashboard

The tutor dashboard provides an overview of the tutor's lessons, students, schedules, and payments. It allows the tutor to manage available lesson times, check student bookings, respond to rescheduling requests, and monitor payment records. This helps the tutor organize lessons and keep track of important information in one place.

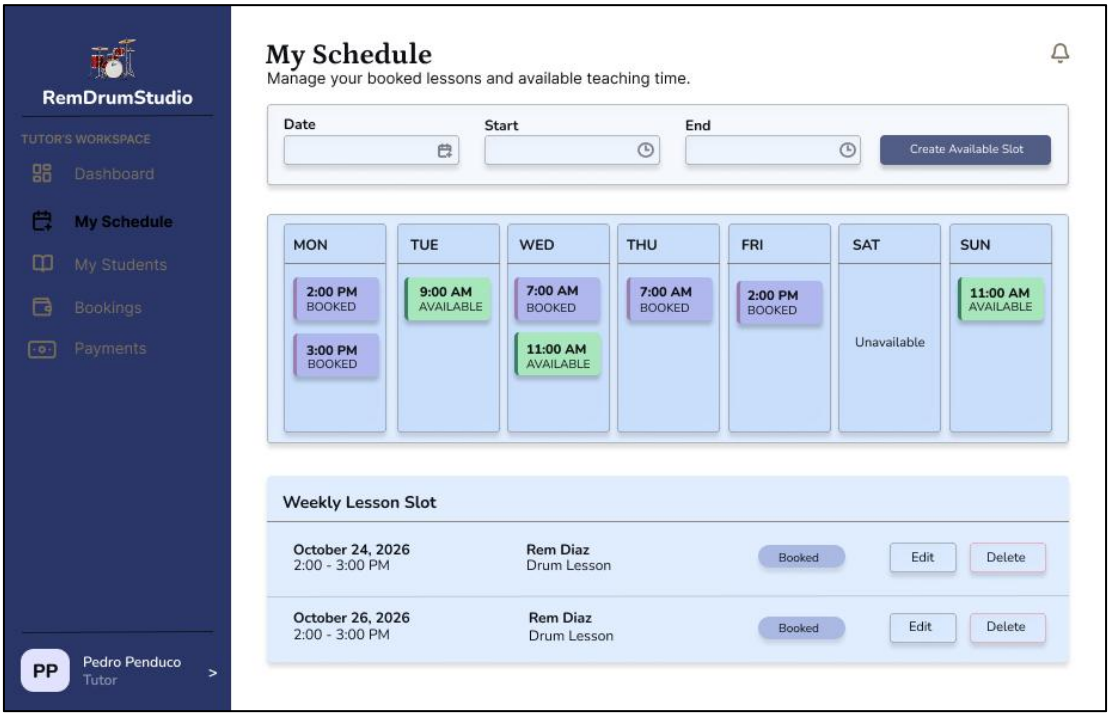


Figure 9. Tutor's Compilation of Schedules

The schedule page allows the tutor to manage available and booked lesson times. The tutor can add new schedules, check which time slots are available, and edit or delete schedules when needed. This helps the tutor organize lessons and avoid schedule conflicts.

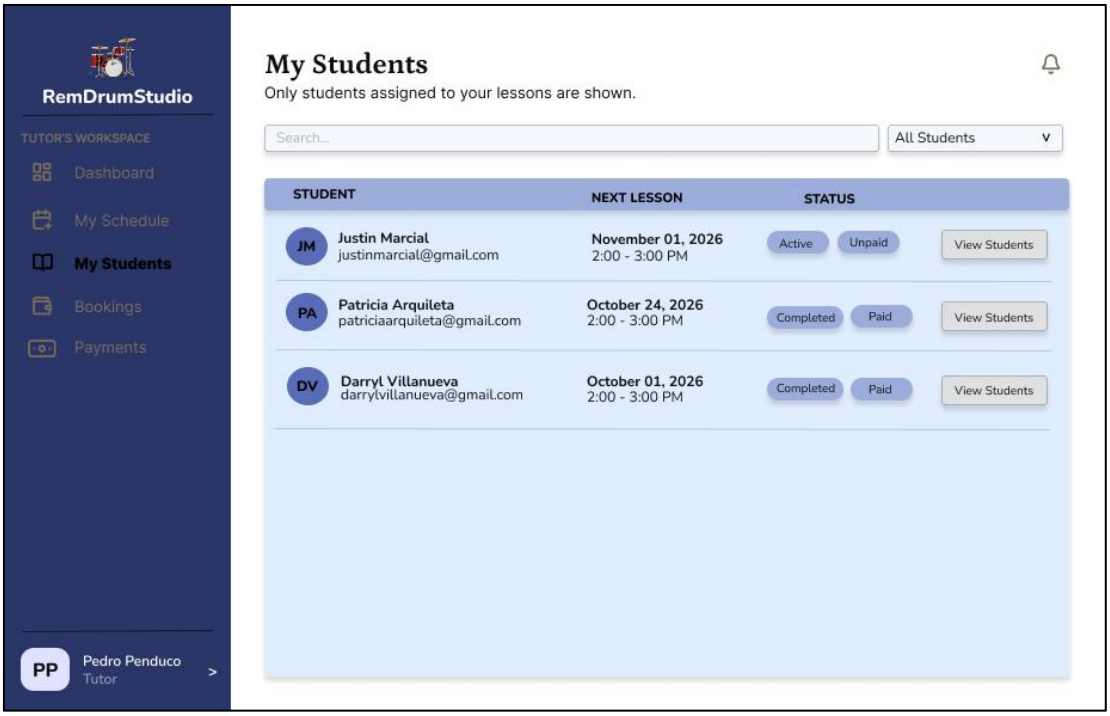


Figure 10. Tutor's List of Students

The student list shows the students registered for lessons with the tutor. It includes their names, email addresses, next lesson schedules, and payment status. The tutor can also search for students and view their details.

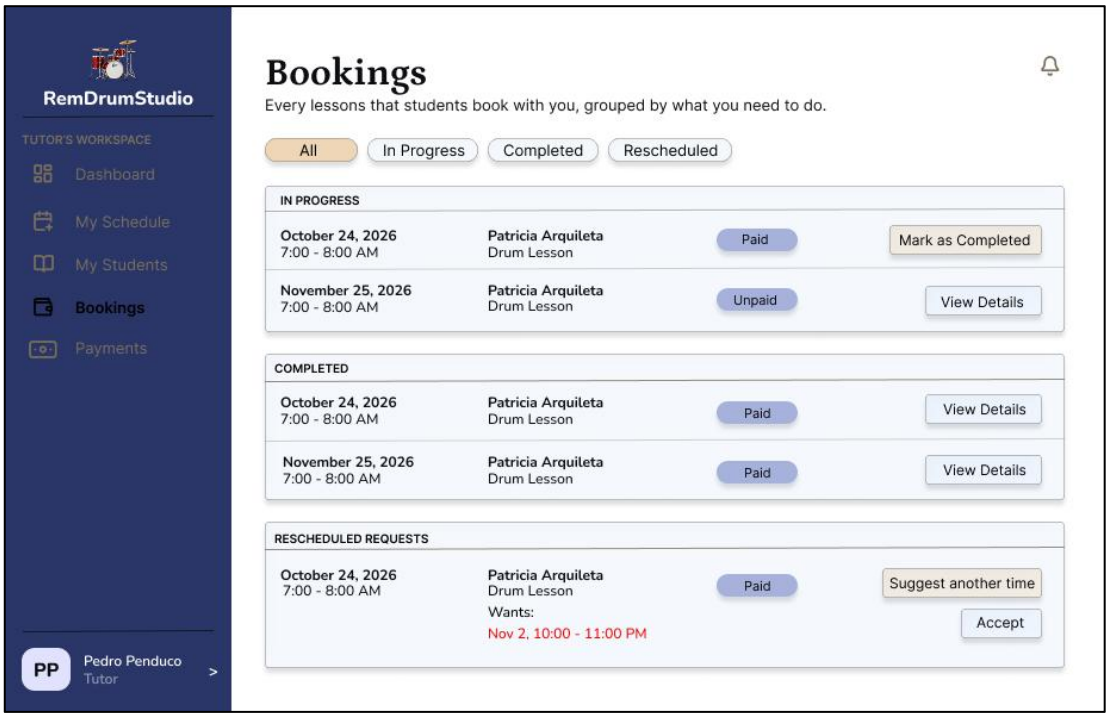


Figure 11. Tutor's Booking Lesson

The bookings page allows the tutor to check all lesson bookings and their current status. The tutor can view ongoing lessons, complete lessons, and reschedule requests. The tutor can also update lesson status and respond to requests for a new schedule.

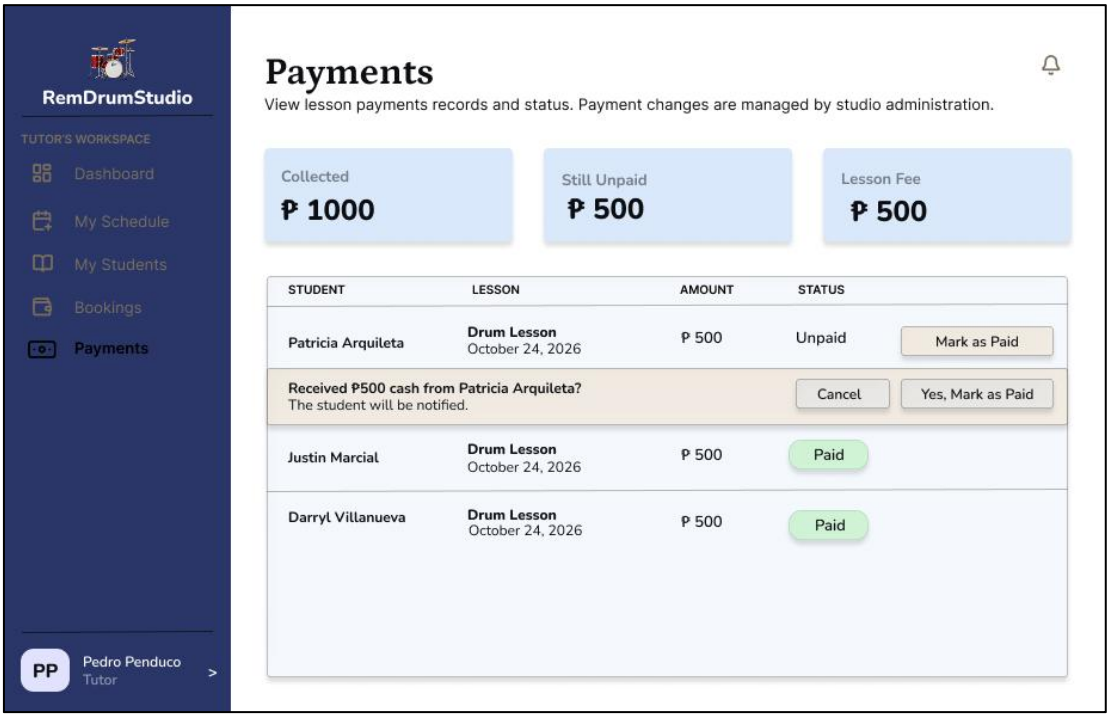


Figure 12. Tutor's Record of Payment

The payment page helps the tutor monitor lesson payments. It shows the total amount collected, unpaid payments, and lesson fees. The tutor can check each student's payment status and mark a lesson as paid after confirming the payment.

C. Student Dashboard

The student dashboard is the workspace of a student in RemDrumStudio. It allows the student to view the next and upcoming lessons, book an available lesson time, request a reschedule, and check lesson fees and payment status. The following pages are available to students.

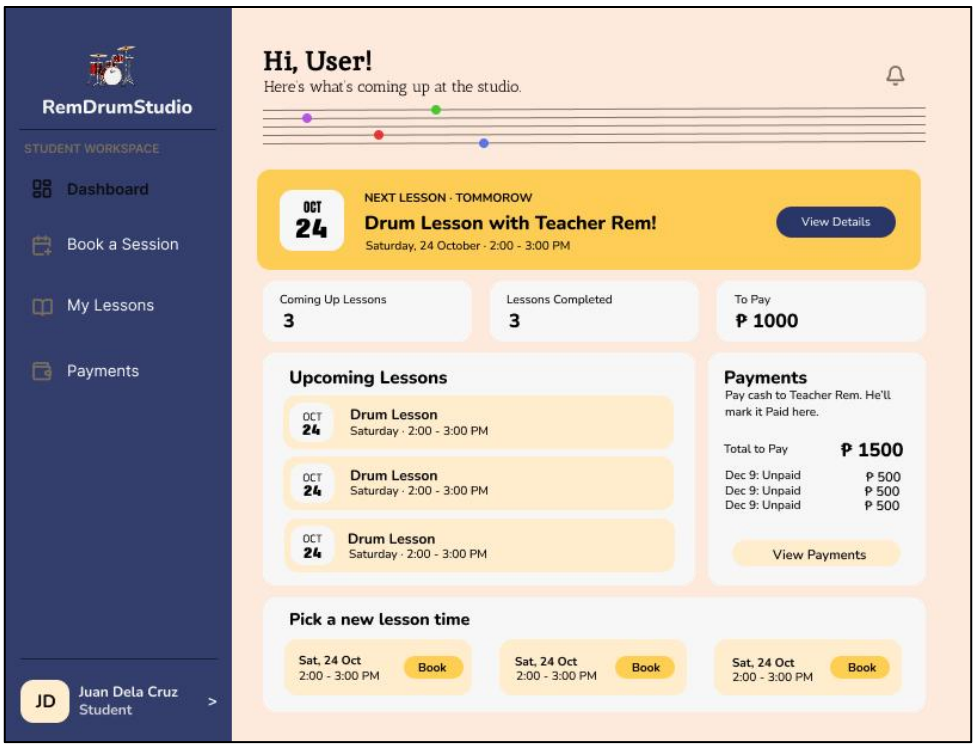


Figure 13. Student Dashboard

The dashboard shows the student's next lesson, upcoming sessions, completed lessons, and payment summary. It also displays available lesson times that the student can book. This helps students check their schedules and payment information in one place.

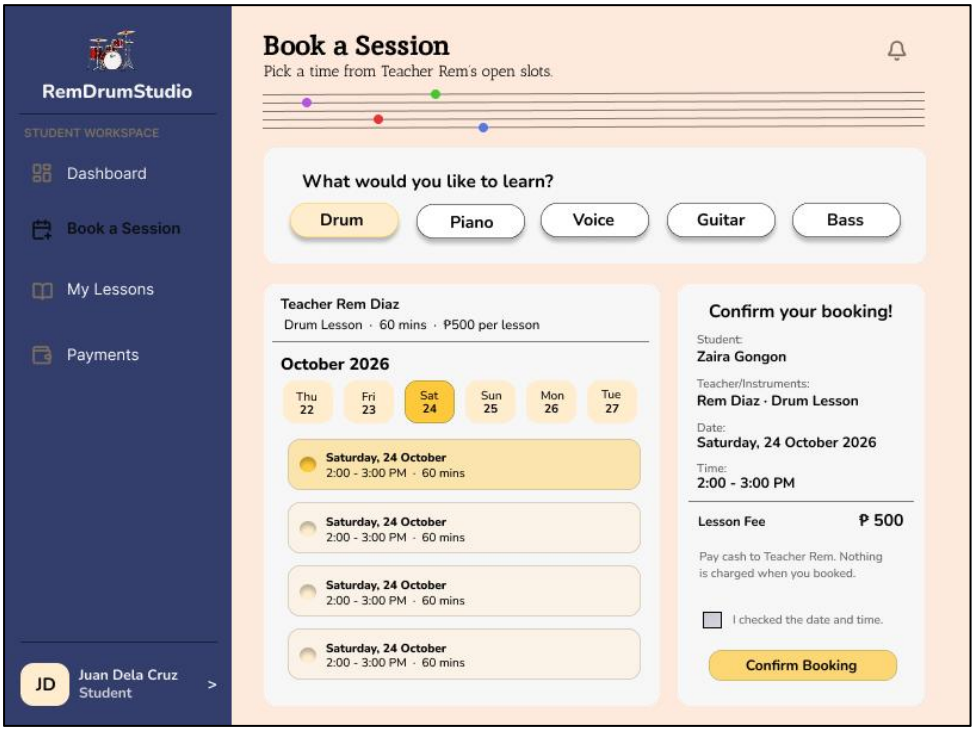


Figure 14. Student's Session Details

My Lessons page allows students to view their booked lessons and check their schedules. It displays lesson details, such as the date, time, tutor, fee, and booking or payment status. Students can also access the option to request a reschedule when needed.

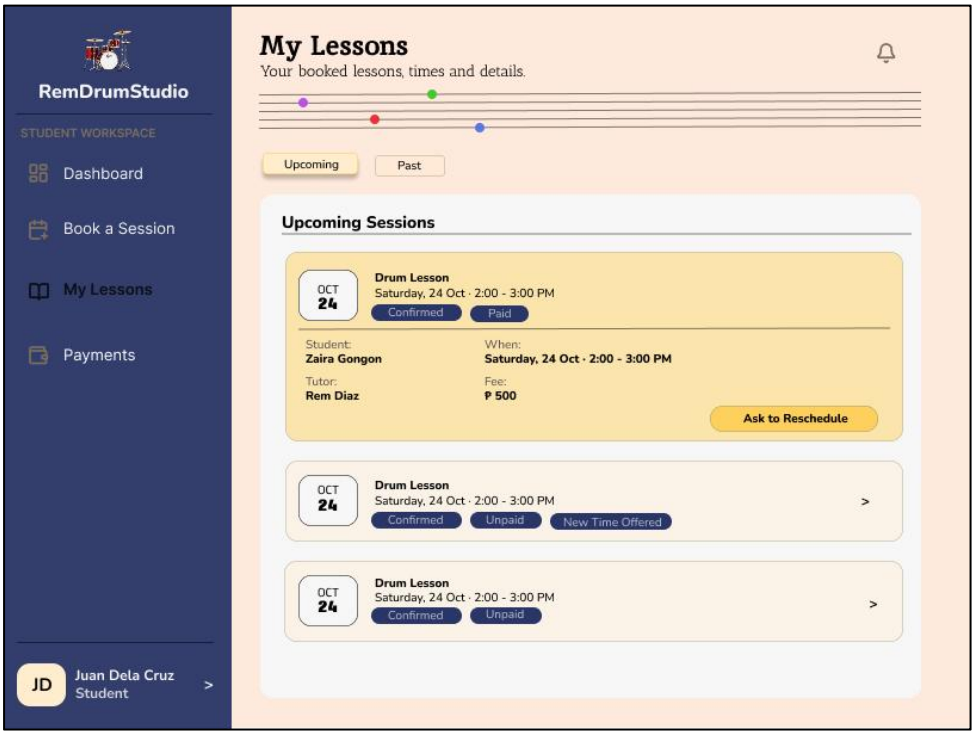


Figure 15. Student Booked Lessons

Figure 15 shows the details of a student's rescheduling request. It displays the original lesson schedule, the requested replacement time, and the reason for the change. It also informs the student that the request is waiting for the tutor's confirmation.

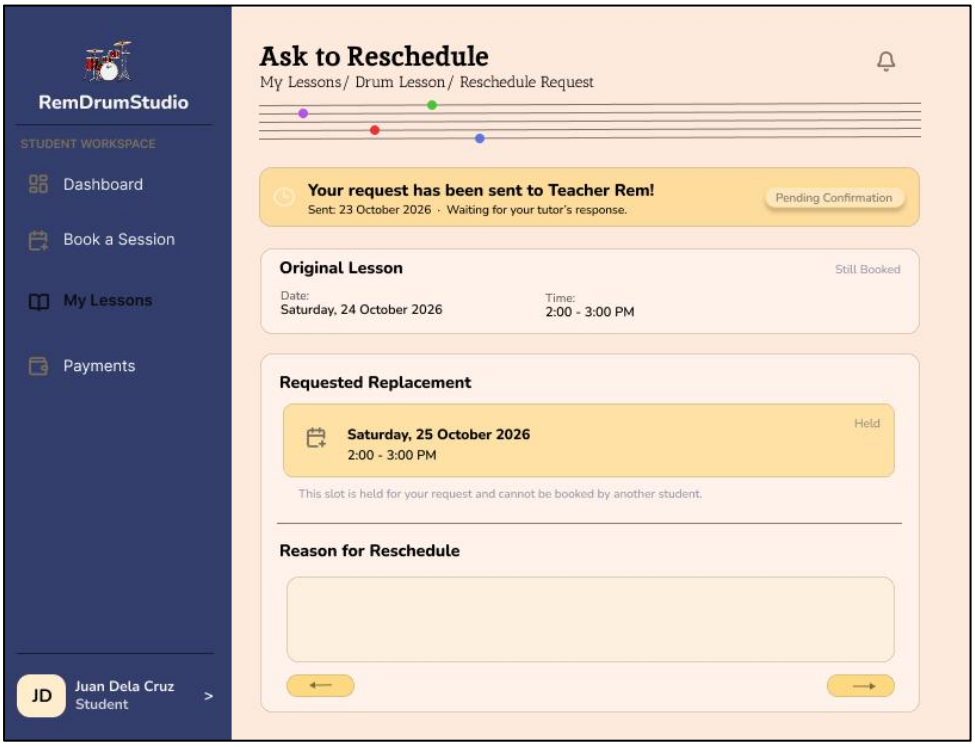


Figure 16. Requesting Reschedule

The rescheduling page allows students to request a change to their booked lesson. The student can review the original schedule, select an available replacement time, and provide a reason for the request. The system holds the requested time while waiting for the tutor's response.

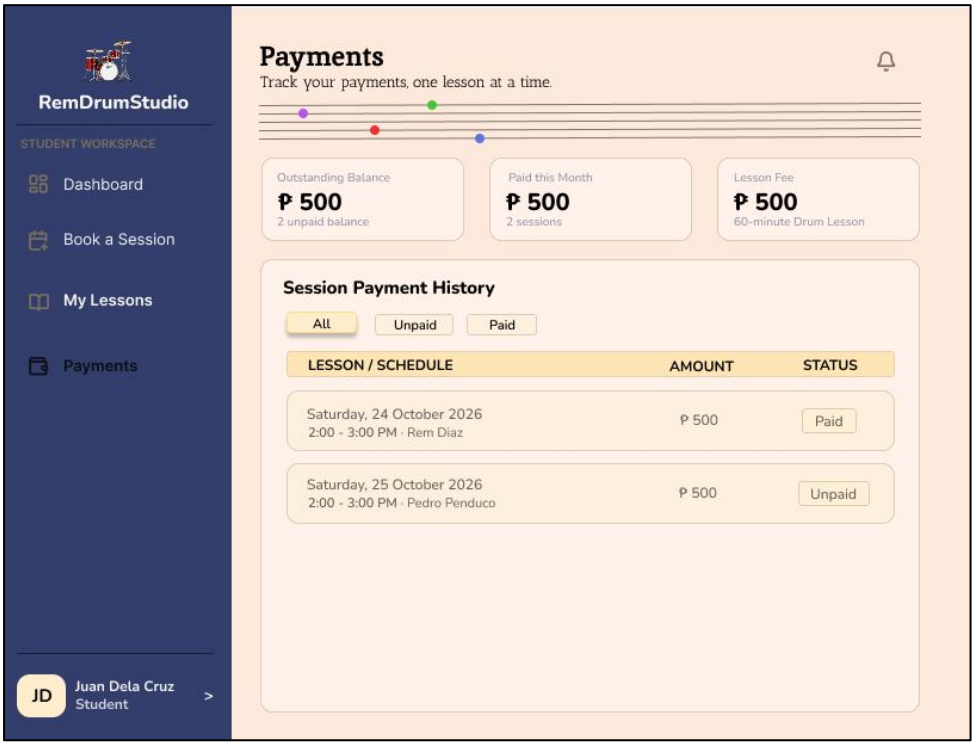


Figure 17. Payments

The payments page allows students to check their lesson fees and payment status. It helps them see which lessons have been paid and which still have unpaid balances. This makes it easier for students to monitor their payments and avoid confusion about their fees.

Exception Handling

This section describes how RemDrumStudio responds when a user's action cannot be completed. The following exception handling is identified for this project's solution.

1. Role-Based Authorization
    a. Shows an error when a student tries to access the tutor's dashboard, or vice versa.
    b. Shows an error when the email or password entered is invalid.
2. Registration and Rescheduling Management
    a. Shows an error when a student tries to reschedule a lesson even though they are not registered.
    b. Shows an error when the selected schedule is already taken.
    c. Shows an error when the student tries to reschedule without having an existing appointment.
    d. Shows an error when the registration details are incomplete or incorrect.
    e. Shows an error when the student tries to reschedule less than 1 day before the session.
    f. Shows an error when the student does not give a reason for rescheduling.
3. Payment Tracking
    a. Shows an error when there is no student or payment information available to display.
    b. Shows an error when the system cannot find the student's payment record.
    c. Shows an error when the payment amount entered is invalid or incomplete.
    d. Shows an error when the payment information cannot be properly recorded or updated.
    e. Shows an error when a session that is already paid is paid again.

Entity Relationship Diagram

The Entity Relationship Diagram, or ERD, shows how the different data in the RemDrumStudio system are organized and connected to each other. It identifies the main entities, their attributes, and the relationships between them. Through this diagram, we can understand how the system manages user accounts, student information, tutor schedules, enrollment, and payments in an organized way.

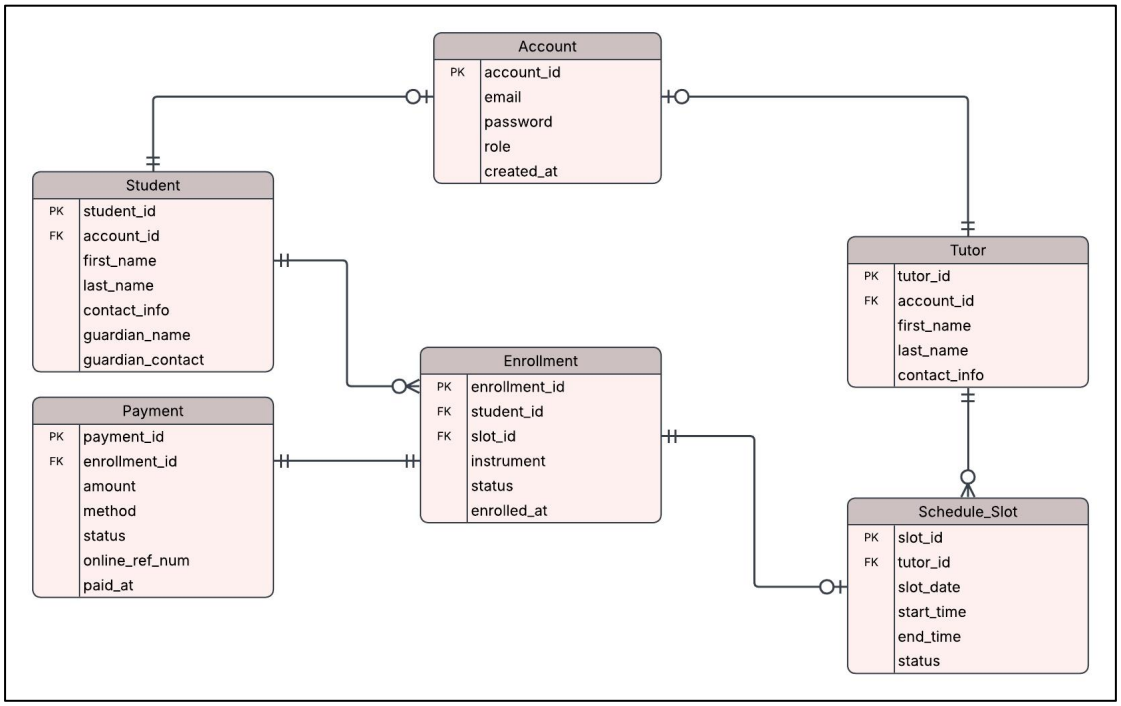


Figure 18. Entity Relationship Diagram of RemDrumStudio

The relationships between the entities in the diagram are described as follows. Every student and tutor profile belongs to exactly one account, which stores email, password, and role used to log in, so an account is linked to at most one profile, either a student or a tutor. A students can have many enrollments, but each enrollment belongs to only one student, and a student who has not booked a lesson yet has no enrollment. A tutor uploads many schedule slots, while each slot belongs to exactly one tutor. An enrollment is made for exactly one slot, and a slot is booked by at most one enrollment, so a slot with no enrollment is still available and two students cannot book the same slot. Finally, each enrollment has one payment record that holds the lesson fee and its paid or unpaid status, and each payment records belong to only one enrollment.

System Architecture Design

The System Architecture Design illustrates the overall structure of the RemDrumStudio system and how its different components work together. It uses a layered architecture, where each layer has a specific function, starting from the users who access the system through their devices up to the database where information is stored. This design helps explain how the system processes user requests, manages schedules and payments, and maintains organized and secure records.

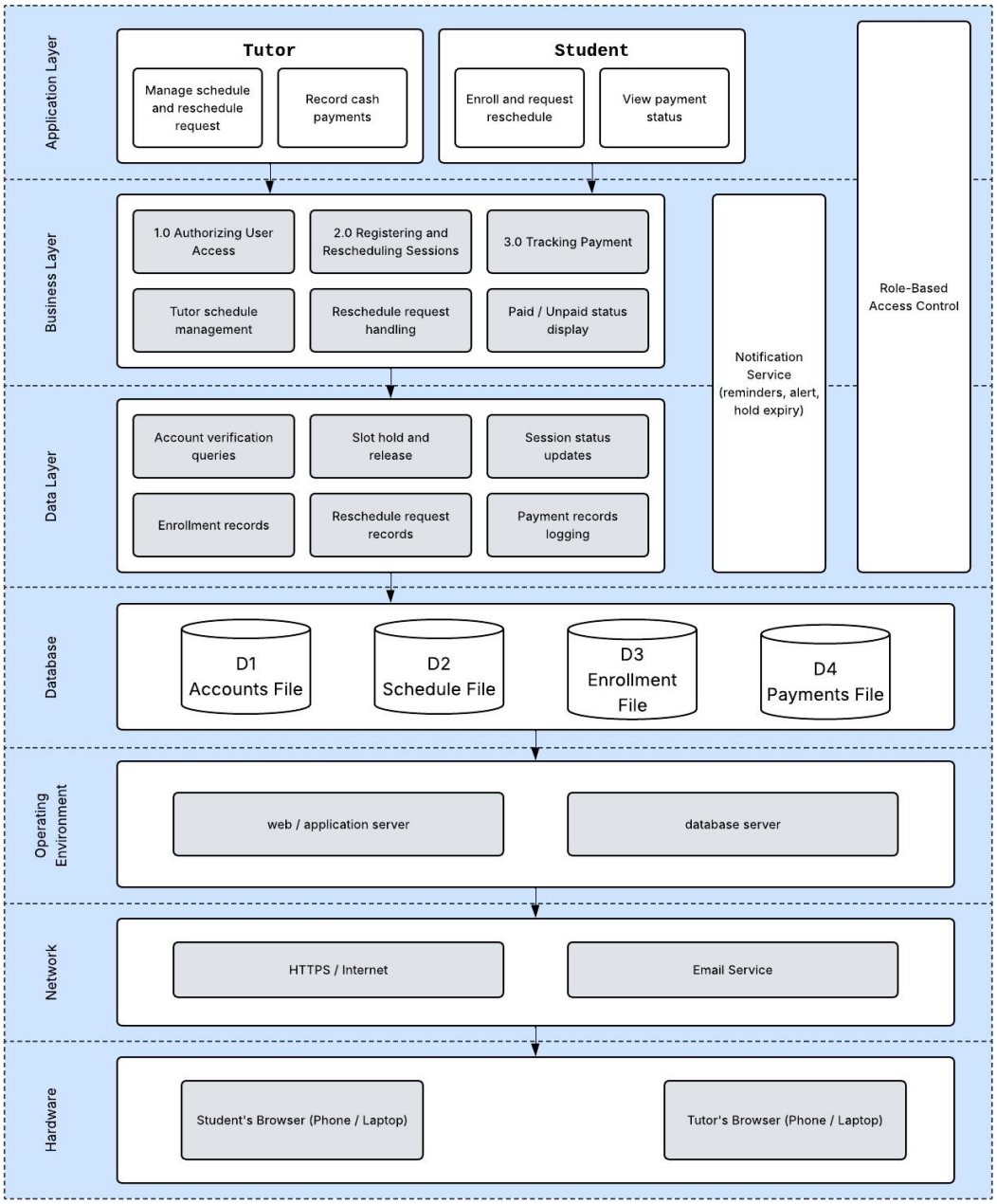


Figure 19. Layered Architecture Diagram of RemDrumStudio

Figure 19 shows how RemDrumStudio is organized into layers and how a request moves through them. The **Application Layer** is where the tutor and the student use the system. The tutor manages the schedule and reschedules requests and records cash payments, while the student enrolls, requests a reschedule, and views the payment status. These requests are passed to the **Business Layer**, which contains the main processes of the system, namely authorizing user access, registering and rescheduling sessions, and tracking payment. It also contains their supporting functions, which are tutor schedule management, reschedule request handling and the display of paid or unpaid status.

In addition, Business Layer relies on the Data Layer to read and write information. **Data Layer** handles account verification, the holding and releasing of slots, session status updates, and the enrollment, reschedule request and payment records. Then, this layer saves and retrieves these records from the **Database**, which consists of the Accounts File, the Schedule File, the Enrollment File and the Payments File. However, there are two services that run alongside the Application, Business and Data Layers. This includes the **Role-Based Access Control** that limits each user to the functions of their role, and the **Notification Service** in which sends reminders and alerts and releases held slots that are not confirmed in time.

The lower layers describe where the system runs and how it is reached. In the **Operating Environment**, the system is hosted on a web or application server and a database server. In the **Network layer**, users connect through the Internet using HTTPS, and the system is linked to the email service used for notifications. At the bottom, the **Hardware** layer shows that the student and the tutor access the system through a web browser on a phone or a laptop.

Test Scenarios

The following are test conditions and expected results for each main functionality of the system.

TestId	Test Condition	Expected Result
1	Student logs in with a registered email, the correct password, and the student role	Student Dashboard is displayed.
2	Tutor logs in with a registered email, the correct password, and the tutor role	Tutor Dashboard is displayed.
3	User logs in with an unregistered email or an incorrect password	User in unable to log in and the error “Invalid email or password.” is shown on the screen
4	User logs in with correct credentials but selects a role that does not match the account	User is denied access and the error “Access denied. This account is not registered as a Student/Tutor.” is shown on the screen
5	New user signs up with complete details and selects a role	Account is created in account_tbl and the dashboard of the selected role is displayed
6	New user signs up with missing details	Account is not created and the error “Please complete all required fields.” is shown on the screen

Table 2. Test Scenarios of RemDrumStudio

Main Processing

The following details describe the system's main processing.

Feature	Algorithm
Role-Based Authorization Page	Input Parameter: action (Log In or Sign Up), email, password, role (Student or Tutor) • Reads input and assign to varAction, varEmail, varPass, varRole • If varAction is Sign Up ○ If a required detail is missing, throw error “Please complete all required fields.” ○ Else, create the account in account_tbl with varEmail, varPass, and varRole. • Else, validate varEmail and varPass against account_tbl ○ If varEmail and varPass do not match an account, throw error “Invalid email or password.” ○ Else is the role of the account is not varRole, throw error “Access denied. This account is not registered as a [varRole].” • If no error was thrown ○ If varRole is Student, go to Student Dashboard ○ Else, go to Tutor Dashboard

Table 3. Main processing of RemDrumStudio